

Node.js Server Deployment on AWS EC2 using PM2

AWS EC2 Node.js 18 PM2 Process Manager

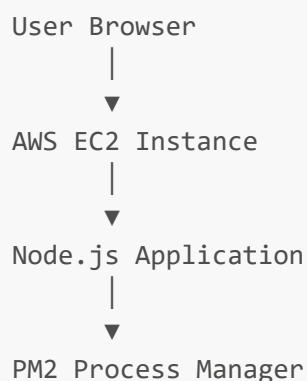
Project Overview

This project demonstrates deployment of a Node.js application on AWS EC2 using:

- AWS EC2
- Node.js
- Express.js
- PM2
- GitHub
- Amazon Linux 2023

The application is publicly accessible using the EC2 Public IP on port 3000.

Architecture



Project Structure

```
node-server-project/
├── app.js
├── package.json
├── README.md
└── screenshots/
```

AWS EC2 Details

Service	Details
Cloud Provider	AWS
Instance Type	t3.micro
Operating System	Amazon Linux 2023
Runtime	Node.js 18
Process Manager	PM2

Step 1 — Launch EC2 Instance

1. Open AWS Console
 2. Go to EC2 Dashboard
 3. Click **Launch Instance**
 4. Select:
 - Amazon Linux 2023
 - t3.micro
 5. Configure Security Group:
 - SSH → Port 22
 - HTTP → Port 80
 - Custom TCP → Port 3000
-

Step 2 — Connect to EC2

```
ssh -i your-key.pem ec2-user@YOUR_PUBLIC_IP
```

Step 3 — Update Server

```
sudo yum update -y
```

Step 4 — Install Node.js

```
sudo yum install nodejs -y
```

Verify Installation

```
node -v  
npm -v
```

Step 5 — Clone GitHub Repository

```
git clone https://github.com/YOUR_USERNAME/node-server-project.git
```

Move into Project Directory

```
cd node-server-project
```

Step 6 — Install Dependencies

```
npm install
```

Step 7 — Install PM2

```
sudo npm install pm2 -g
```

Step 8 — Start Application using PM2

```
pm2 start app.js
```

Enable Auto Restart on Server Reboot

```
pm2 startup  
pm2 save
```

Check Running Applications

```
pm2 list
```

Access Application

Open browser:

```
http://YOUR_PUBLIC_IP:3000
```

Example:

```
http://52.201.252.160:3000
```

Application Output

```
Node Server  
Hello from AWS EC2 + Node.js + PM2  
CI/CD Deployment Successful
```

Screenshots

AWS EC2 Dashboard

Instances (1/3) Info

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	node-Server	i-0d2c1aade2c54a64e	Running	t3.micro	3/3 checks passec	us-east-1c	ec2-52-201-252-160.co...
☐	my_portfolio	i-070318c4af7b427ab	Stopped	t3.micro	-	us-east-1d	-

i-0d2c1aade2c54a64e (node-Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary Info

Instance ID

i-0d2c1aade2c54a64e

Public IPv4 address

52.201.252.160 | open address

Private IPv4 addresses

Install Node.js

```
~/my/
[ec2-user@ip-172-31-29-137 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-29-137 ~]$ sudo yum install nodejs -y
Last metadata expiration check: 0:00:19 ago on Sat May 16 07:36:58 2026.
Dependencies resolved.

=====
Package                                Architecture           Version
=====
Installing:
nodejs                                 x86_64                  1:18.20.8-1.amzn2023.0.2
Installing dependencies:
libbrotli                             x86_64                  1.0.9-4.amzn2023.0.2
nodejs-libs                           x86_64                  1:18.20.8-1.amzn2023.0.2
Installing weak dependencies:
nodejs-docs                           noarch                  1:18.20.8-1.amzn2023.0.2
nodejs-full-i18n                      x86_64                  1:18.20.8-1.amzn2023.0.2
nodejs-npm                            x86_64                  1:10.8.2-1.18.20.8.1.amzn2023.0.2
=====

Transaction Summary
-----
Install 6 Packages

Total download size: 45 M
Installed size: 224 M
Downloading Packages:
(1/6): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm
(2/6): nodejs-docs-18.20.8-1.amzn2023.0.2.noarch.rpm
(3/6): nodejs-full-i18n-18.20.8-1.amzn2023.0.2.x86_64.rpm
(4/6): nodejs-18.20.8-1.amzn2023.0.2.x86_64.rpm
(5/6): nodejs-npm-10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64.rpm
(6/6): nodejs-libs-18.20.8-1.amzn2023.0.2.x86_64.rpm
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Installing : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
Installing : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
Installing : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
Installing : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
Running scriptlet: nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
INFO: registered node-18 in the alternatives

Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Verifying : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
Verifying : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
Verifying : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
Verifying : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
Verifying : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
```

GitHub Clone & NPM Install

```
Complete!
[ec2-user@ip-172-31-29-137 ~]$ git clone https://github.com/iamtruptimane/node-js-app-CICD.git
Cloning into 'node-js-app-CICD'...
remote: Enumerating objects: 49, done.
remote: Counting objects: 100% (31/31), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 49 (delta 20), reused 21 (delta 13), pack-reused 18 (from 1)
Receiving objects: 100% (49/49), 5.54 KiB | 5.54 MiB/s, done.
Resolving deltas: 100% (24/24), done.
[ec2-user@ip-172-31-29-137 ~]$ ls
node-js-app-CICD
[ec2-user@ip-172-31-29-137 ~]$ mv node-js-app-CICD/ node
[ec2-user@ip-172-31-29-137 ~]$ ls
node
[ec2-user@ip-172-31-29-137 ~]$ cd/ node
-bash: cd/: No such file or directory
[ec2-user@ip-172-31-29-137 ~]$ cd /node
-bash: cd: /node: No such file or directory
[ec2-user@ip-172-31-29-137 ~]$ cd node
[ec2-user@ip-172-31-29-137 node]$ sudo npm install
```

PM2 Running Application

To go further checkout:
<http://pm2.io/>

```
[PM2] Spawning PM2 daemon with pm2_home=/root/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /home/ec2-user/node/app.js in fork_mode (1 instance)
[PM2] Done.
```

id	name	mode	v	status	cpu	memory
0	app	fork	0	online	0%	37.5mb

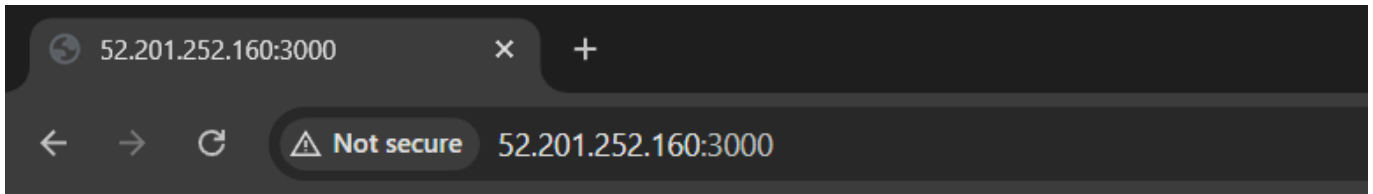


PM2 is a Production Process Manager for Node.js applications with a built-in Load Balancer.

To go further checkout:
<http://pm2.io/>

```
[PM2] Spawning PM2 daemon with pm2_home=/home/ec2-user/.pm2
[PM2] PM2 Successfully daemonized
Use --update-env to update environment variables
```

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Hello from jenkins, added webhook, we are from 14 july batch

Useful PM2 Commands

Restart Application

```
pm2 restart app.js
```

Stop Application


```
pm2 stop app.js
```

Delete Application

```
pm2 delete app
```

View Logs

```
pm2 logs
```

Security Group Configuration

Type	Port
SSH	22
HTTP	80
Custom TCP	3000

Future Improvements

- Jenkins CI/CD Pipeline
 - Nginx Reverse Proxy
 - Docker Containerization
 - HTTPS SSL Setup
 - Application Load Balancer
 - Auto Scaling Group
 - Route53 Domain Configuration
-

Author

Nilesh Pradeep Patil

Fortune Cloud Technology
MCA AWS & DevOps Engineering

summary

Successfully deployed a Node.js application on AWS EC2 using PM2 and GitHub integration. The server is publicly accessible and production-ready for future CI/CD implementations.
